SUMMARISATION AI

A Project Report

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Abstract

This project presents an AI-based text summarisation system built using Transformer-based deep learning models. The system uses the **BART-large-CNN** pre-trained model to generate concise, context-aware summaries of large documents. By leveraging HuggingFace's Transformers pipeline, the system performs abstractive summarisation—creating new sentences rather than copying text. The project automates summarisation tasks using minimal code while ensuring high-quality results suitable for academic, news, and professional content. The solution demonstrates the efficiency and real-world applicability of modern NLP models for condensing long text into meaningful summaries.
Keywords: **NLP, Transformers, Summarisation, BART, HuggingFace, Deep Learning**.

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1. Introduction

1.1 Overview

With rapid digitalisation, the amount of textual data across the internet, news, research, and documents has grown exponentially. Manual summarisation of large text is time-consuming and prone to human error. AI-based text summarisation provides an efficient solution by automatically generating short, meaningful summaries of long documents.

This project uses the **BART-Large-CNN model**, a state-of-the-art transformer designed specifically for summarisation tasks. The summarisation workflow leverages HuggingFace's Transformer library, offering fast and accurate results.

1.2 Objectives

- To build an AI-powered text summarisation tool using Transformer models.
- To implement abstractive summarisation with a pre-trained BART model.
- To automate summary generation for long articles, research texts, and documents.
- To evaluate the accuracy and efficiency of modern NLP summarisation techniques.
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1.3 Problem Statement

The vast growth of digital information makes manual summarisation impractical. Traditional methods fail to capture context, coherence, and meaningful structure. There is a need for an automated system that can accurately summarise long documents while preserving core information.

2. Literature Review

Transformer-based architectures such as BERT, GPT, T5, and BART have reshaped NLP tasks by enabling parallel processing and deep contextual understanding.
The **BART-Large-CNN** model has been fine-tuned using the CNN/DailyMail dataset containing **300,000+ news articles with human-written summaries**, making it highly effective for summarisation.

Studies demonstrate that pre-trained models outperform traditional extractive methods, producing human-like summaries with better coherence and contextual accuracy.

3. Methodology

3.1 The overall architecture of the Summarisation AI system includes:

3.2 Input text file/document

3.3 Pre-trained BART transformer model

3.4 Tokenization & encoding pipeline

3.5 Text generation engine for summarisation

3.6 Output summary file

3.7 This simple yet powerful pipeline allows high-performance NLP with minimal code.

3.8 Hardware Components

3.9 Software Tools

4. Implementation

- Input text is tokenized.
- Transformer encoder-decoder architecture processes the sequence.
- Model generates a shorter paraphrased version.
- Output is written to a file.

5. Results and Discussion

The summarisation system successfully condenses long paragraphs into short, meaningful summaries.

5.1 Observed Benefits

- **High accuracy:** BART-large-cnn produces coherent human-like summaries.
- **Fast inference:** Generates summaries in seconds.
- **Minimal code:** Less than 10 lines needed for end-to-end summarisation.

5.2 Challenges

- Very long text requires chunking.
- Output quality depends on model training domain (news dataset).
- Occasional loss of specific details during abstractive generation

6. Conclusion and Future Scope

6.1 Conclusion

This project demonstrates the efficiency of Transformer-based models for automated text summarisation. The use of a pre-trained BART summarisation model eliminates the need for expensive training while delivering high-quality summarisation results. The system is simple, scalable, and can be integrated into a wide range of applications.

6.2 Future Scope

- I Building a **GUI-based summarisation application**.
- Deploying the model as a **web API** or mobile app.
- Fine-tuning BART for specific domains (medical, legal, education).
- Adding multilingual summarisation support.
- Integrating speech-to-text to summarise audio/video content.

7. References

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3. HuggingFace Transformers Library Documentation.

4. CNN/DailyMail Dataset used for model fine-tuning (Referenced in PPT).

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